

Deployment Standard Operating Procedure

This SOP standardizes the detailed steps, checks, approvals, and evidence required to perform controlled production deployments.

Department: Sop | Version: 1.1 | Owner: Michele Williams, Release Manager

Access Level: Internal - Operations and Engineering | Created: 2025-11-25 | Updated: 2026-02-03 | Source File: sop_deployment_v1.pdf

1.0 Purpose

This procedure exists to make production deployments repeatable, observable, and audit-ready. A production-level RAG dataset should include not just policy language, but the exact steps an operator follows before, during, and after a release so the retriever can surface execution detail under real pressure.

The SOP bridges engineering preparation with operational control. It defines a step sequence that reduces ambiguity, clarifies who is expected to act at each stage, and ensures the release record captures enough evidence to support review, rollback decisions, and future improvement work.

2.0 Scope

This procedure applies to planned production deployments for services and systems that support customer-facing or business-critical workflows. Team-specific tooling may vary, but the control sequence in this SOP is the expected baseline unless an approved platform exception exists.

The SOP covers pre-deployment checks, deployment execution, validation, communication, and closure. It should be used together with service-specific runbooks, migration notes, and monitoring dashboards when a real release is performed.

- Applies to standard releases, high-risk changes, and progressive rollouts.
- Requires named ownership, evidence capture, and rollback preparedness before execution.
- Must be followed even when automation performs the technical release steps.

3.0 Responsibilities

The responsibilities below define who must act, approve, review, and maintain records so the process can be executed consistently across teams and audit periods.

3.1 Release Manager

- Own the checklist, confirm readiness evidence, and coordinate the release timeline.
- Ensure the right approvers, responders, and communication channels are in place before start.
- Close the release only after validation and documentation requirements are completed.

3.2 Implementing Engineer

- Provide the release artifact, migration notes, rollback guidance, and expected health indicators.
- Execute technical steps or supervise automation behavior during the release.
- Call out unexpected signals immediately rather than waiting for certainty.

3.3 Operational Support

- Monitor dashboards, confirm incident readiness, and assist with rollback or escalation if needed.
- Record time-stamped actions and communications during the release window.
- Verify closure evidence is stored in the release record.

4.0 Pre-Deployment Procedure

Operators should begin by confirming that the planned release matches the approved scope and that every prerequisite item is present before any environment change starts. This step is deliberately explicit because incomplete preparation is a common cause of rushed release decisions and weak rollback outcomes.

4.1 Step 4.1 Confirm Scope and Approval

Review the release ticket, linked pull requests, and any migration notes to confirm the exact change set. Verify that the named approver, release owner, and technical operator are correct and that the planned window still matches business risk and dependency timing.

- Check the approved artifact or branch.
- Verify rollout window, approver, and owner assignments.
- Confirm that non-release changes are not bundled into the same deployment unexpectedly.

4.2 Step 4.2 Confirm Readiness Evidence

Validate test completion, lower-environment results, dashboard links, and rollback instructions. If a migration, flag, or configuration change is involved, confirm sequencing and identify the last safe rollback point before execution begins.

- Review automated and manual validation results.
- Confirm dashboards, alerts, and health checks are available.
- Verify rollback conditions and any irreversible steps.

5.0 Deployment Execution Procedure

During deployment, the operator should work from the checklist in order and capture actions as they occur. Even when the rollout is automated, the human operator remains responsible for validating system behavior and pausing when evidence suggests the change is unsafe to continue.

5.1 Step 5.1 Start the Deployment

Trigger the deployment from the approved artifact, record the start time, and announce the beginning of the release in the designated channel. This creates a shared reference point for downstream responders and provides the basis for later timeline reconstruction.

- Start the rollout through the standard pipeline.
- Record operator name, start time, and artifact version.
- Notify stakeholders that deployment is in progress.

5.2 Step 5.2 Observe Each Rollout Stage

As the deployment moves through stages, review health indicators rather than assuming the stage is successful because the pipeline advanced. For progressive rollouts, validate early signals before expanding traffic or promoting the release more broadly.

- Check build status, migration output, and environment health after each stage.
- Pause before broad exposure if error trends, auth failures, or latency regressions appear.
- Use the incident route immediately if service health degrades materially.

5.3 Step 5.3 Validate Business-Critical Flows

Run the agreed validation set for the service, including the user journeys or operational workflows most likely to reveal a bad release. Validation should focus on the behaviors that matter to customers or internal operators, not only generic service health.

- Check critical read, write, auth, and background-processing paths.
- Compare live metrics to expected release behavior.
- Document any known issue or anomaly before deciding to continue.

6.0 Post-Deployment and Rollback Procedure

Once the rollout completes, the operator should remain in a monitoring posture until the agreed stabilization window ends. If meaningful customer impact or data risk appears, the release manager should move directly to rollback or a documented mitigation path rather than letting uncertainty delay the response.

6.1 Step 6.1 Stabilization Monitoring

- Watch dashboards, alerts, logs, and traces for the agreed observation period.
- Confirm that queue depth, background jobs, and dependent systems remain healthy.
- Communicate status updates at the cadence defined for the release window.

6.2 Step 6.2 Rollback Decision

If health indicators or business outcomes show the release is causing harm, execute the rollback according to the documented plan. The decision should prioritize service stability and data safety over optimism that the issue may self-correct.

- Declare rollback when thresholds or user impact criteria are met.
- Record the rollback trigger, time, and the systems affected.
- Return to monitoring after rollback until the system is stable.

6.3 Step 6.3 Release Closure

- Post the final status, known issues, and follow-up tasks in the release record.
- Attach evidence, timeline notes, and any deviations from the planned checklist.
- Close the release only after documentation and communication are complete.

7.0 Exceptions and Escalation

Any skipped step in this SOP must be documented as an exception with a reason, approver, and compensating control. Operators should never rely on memory to justify why a standard step was omitted during a production release.

Escalate to incident management immediately if the deployment threatens availability, integrity, or customer trust. The SOP is designed to support decisive escalation, not to delay it.

8.0 Records and Review

The release ticket must contain scope, approvals, timestamps, validation evidence, rollback actions if any, and the final release summary. Operational notes should be durable enough that another team can reconstruct the event without private chat history.

Release management reviews deployment adherence, rollback rates, and repeated checklist misses after each major cycle. The SOP should be updated when the same failure mode appears more than once or when automation changes the practical sequence of work.